



LINEAR ALGEBRA AND ITS APPLICATIONS

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Unit-4

Orthogonalization , Eigen Values and Eigen Vectors

Topics in the Module:



- ❖ Orthogonal Bases, orthogonal matrices and properties
- ❖ Rectangular matrices with orthonormal columns
- ❖ The Gram- Schmidt Orthogonalization
- ❖ $A=QR$ factorization
- ❖ Introduction to Eigenvalues and Eigenvectors
- ❖ Properties of Eigenvalues and Eigenvectors and C-H theorem
- ❖ Problems on Eigenvalues and Eigenvectors and C-H theorem
- ❖ Symmetric Matrices and Diagonalization of a Matrix
- ❖ Problems on diagonalization of a matrix
- ❖ Powers and products of matrices

CLASS-1

ORTHOGONAL BASES, ORTHOGONAL MATRICES AND PROPERTIES

In an orthogonal basis, every vector is perpendicular to every other vector.

The coordinate axes are mutually orthogonal.

Mutually perpendicular unit vectors are called Orthonormal vectors.

For the vector space $\mathbb{R}^2$,

1. The set $(2, 0)$, $(0, 2)$ is an orthogonal basis.
2. The set $(1, -2)$, $(2, 1)$ is an orthogonal basis.
3. The set $(1, 0)$, $(0, 1)$ is an orthonormal basis.

- A matrix with Orthonormal columns will be called Q .
- A square matrix with Orthonormal columns is called an **Orthogonal matrix** denoted by Q .

Ex: Rotation matrix , any permutation matrix .

Properties of Q

- If Q (square or rectangular) has orthonormal columns, then $Q^T Q = I$.
- An orthogonal matrix is a square matrix with orthonormal columns. Then Q^T is Q^{-1} .
- If Q is rectangular then Q^T is **left inverse** of Q .
- Multiplication by any Q preserves length. The norms of x and Qx are equal.

- Also, Q preserves inner products and angles, since $(Qx)^T(Qy) = x^TQ^TQy = x^Ty$.

If $q_1, q_2, \dots, q_n$ are orthonormal basis of R^n then any vector b from R^n can be expressed as

$$b = x_1q_1 + x_2q_2 + \dots + x_nq_n \quad \dots \text{Eqn (1)}$$

Multiply both sides by q_1^T . Then $q_1^Tb = x_1$.

Similarly, $x_2 = q_2^Tb, \dots, x_n = q_n^Tb$.

Hence, $b = (q_1^Tb)q_1 + (q_2^Tb)q_2 + \dots + (q_n^Tb)q_n$
= sum of one dimensional projections on to q_i 's.

The matrix form of equation (1) is $Qx = b$ and the solution of this system of equations is

$$x = Q^{-1}b = Q^Tb$$

- The rows of a square matrix are orthonormal whenever the columns are

Orthonormal columns

Orthonormal rows

$$Q = \begin{bmatrix} 1/\sqrt{3} & 1/\sqrt{2} & 1/\sqrt{6} \\ 1/\sqrt{3} & 0 & -2/\sqrt{6} \\ 1/\sqrt{3} & -1/\sqrt{2} & 1/\sqrt{6} \end{bmatrix}.$$



THANK YOU

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